

Properties of Real-Valued Functions

August 12, 2025

Key Definitions

Definition (Finite Set)

A set D is **finite** if:

- $D = \emptyset$, or
- $\exists n \in \mathbb{N}$ and a bijection $f : \{1, \dots, n\} \rightarrow D$.

Otherwise, D is **infinite**.

Definition (Countable Set)

A set D is **countable** if:

- It is finite, or
- There exists a bijection $f : \mathbb{N} \rightarrow D$ (countably infinite/denumerable).

A set is **uncountable** if it is not countable.

Countably Infinite Sets:

- $\mathbb{Z}$: The integers.

$$\text{Bijection: } f(n) = \begin{cases} n/2 & \text{if } n \text{ even,} \\ -(n+1)/2 & \text{if } n \text{ odd.} \end{cases}$$

- $\mathbb{Q}$: The rationals (via Proposition 1.14).

Uncountable Sets:

- $\mathbb{R}$: The real numbers (Cantor's diagonal argument).
- $\mathbb{R} \setminus \mathbb{Q}$: The irrationals.

Proposition 1.14: Countable Union of Countable Sets

Theorem

If $\{A_n : n \in \mathbb{N}\}$ is a family of countable sets, then $\bigcup_{n \in \mathbb{N}} A_n$ is countable.

Proof Sketch (Disjoint Case):

- Enumerate each A_n as $\{a_{n,1}, a_{n,2}, \dots\}$.
- Arrange elements in a grid and traverse diagonally:

$$a_{1,1}, a_{1,2} \rightarrow a_{2,1}, a_{1,3} \rightarrow a_{2,2} \rightarrow a_{3,1}, \dots$$

- This creates a bijection $\mathbb{N} \rightarrow \bigcup_n A_n$.

Visualizing the Diagonal Argument

$a_{1,1}, a_{1,2}, a_{1,3}, a_{1,4}$
 $a_{2,1}, a_{2,2}, a_{2,3}, a_{2,4}$
 $a_{3,1}, a_{3,2}, a_{3,3}, a_{3,4}$
 $a_{4,1}, a_{4,2}, a_{4,3}, a_{4,4}$

Diagonal enumeration ensures all elements are covered.

Corollary 1.15: $\mathbb{Q}$ is Countable

Proof.

For each $n \in \mathbb{N}$, define:

$$A_n = \left\{ \frac{m}{n} : m \in \mathbb{Z} \right\}.$$

- Each A_n is countable (bijection $m \mapsto \frac{m}{n}$).
- $\mathbb{Q} = \bigcup_{n \in \mathbb{N}} A_n$.
- By Proposition 1.14, $\mathbb{Q}$ is countable.



Bounded Functions

Definition

Let D be a set and $f : D \rightarrow \mathbb{R}$ be a function.

- f is **bounded above** on D if $\exists \alpha \in \mathbb{R}$ such that $f(x) \leq \alpha$ for all $x \in D$.
- f is **bounded below** on D if $\exists \beta \in \mathbb{R}$ such that $f(x) \geq \beta$ for all $x \in D$.
- f is **bounded** on D if it is bounded above and below.

Note: f is bounded on $D \iff \exists \gamma \in \mathbb{R}$ such that $|f(x)| \leq \gamma$ for all $x \in D$.

Attaining Bounds

Definition

Let $f : D \rightarrow \mathbb{R}$ be a function.

- f **attains its upper bound** if $\exists c \in D$ such that $\sup\{f(x) : x \in D\} = f(c)$.
- f **attains its lower bound** if $\exists d \in D$ such that $\inf\{f(x) : x \in D\} = f(d)$.
- f **attains its bounds** if it attains both.

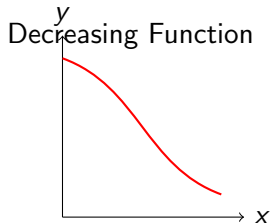
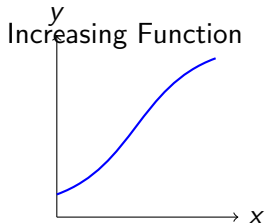
Example: $f(x) = 1 - \frac{x}{n}$: $x \in \mathbb{R}$ with $x > 0$. The supremum is 1 (not attained), and the infimum is $-\infty$ (not bounded below).

Monotonic Functions

Definition

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \rightarrow \mathbb{R}$.

- f is **increasing** on I if $x_1 < x_2 \implies f(x_1) \leq f(x_2)$.
- f is **decreasing** on I if $x_1 < x_2 \implies f(x_1) \geq f(x_2)$.
- f is **monotonic** if it is increasing or decreasing.

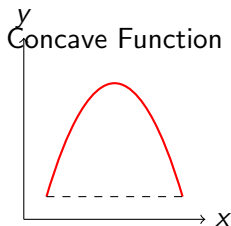
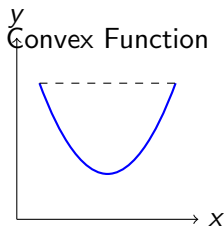


Convex and Concave Functions

Definition

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \rightarrow \mathbb{R}$.

- f is **convex** if the line segment joining any two points on its graph lies on or above the graph.
- f is **concave** if the line segment lies on or below the graph.

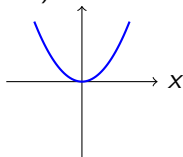


Examples

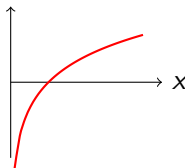
Example 1: $f(x) = x^2$ is convex on $\mathbb{R}$. **Example 2:** $f(x) = \ln(x)$ is concave on $(0, \infty)$.

Non-Example: $f(x) = x^3$ is neither convex nor concave on $\mathbb{R}$, but it is convex on $[0, \infty)$ and concave on $(-\infty, 0]$.

$f(x) = x^2$ (Convex) y



$f(x) = \ln(x)$ (Concave) y



Convexity and Concavity

Definition

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \rightarrow \mathbb{R}$.

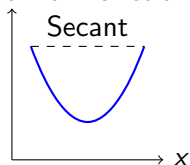
- f is **convex** (concave upward) on I if for all $x_1, x_2 \in I$ and $t \in (0, 1)$:

$$f(tx_1 + (1 - t)x_2) \leq tf(x_1) + (1 - t)f(x_2).$$

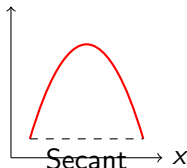
- f is **concave** (concave downward) if:

$$f(tx_1 + (1 - t)x_2) \geq tf(x_1) + (1 - t)f(x_2).$$

Convex Function



Concave Function



Alternative Formulation

Equivalent Definition: For $x_1 < x < x_2$ in I ,

- f is convex if:

$$f(x) \leq f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1).$$

- f is concave if:

$$f(x) \geq f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1).$$

Geometric Interpretation: Convexity means the graph lies **below** secant lines; concavity means it lies **above**.

Convex Functions:

- $f(x) = x^2$ on $\mathbb{R}$.
- $f(x) = e^x$ on $\mathbb{R}$.

Concave Functions:

- $f(x) = \ln(x)$ on $(0, \infty)$.
- $f(x) = \sqrt{x}$ on $[0, \infty)$.

Neither:

- $f(x) = x^3$ on $\mathbb{R}$ (but convex on $[0, \infty)$, concave on $(-\infty, 0]$).

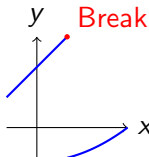
Intermediate Value Property (IVP)

Definition

A function $f : D \rightarrow \mathbb{R}$ has the **IVP** on an interval $I \subseteq D$ if for all $a, b \in I$ and r between $f(a)$ and $f(b)$, there exists $x \in [a, b]$ such that $f(x) = r$.

Examples:

- $f(x) = 2x + 1$ has IVP on $\mathbb{R}$.
- $f(x) = \begin{cases} x + 2 & \text{if } x \leq 1, \\ (x^2 - 9)/8 & \text{if } x > 1, \end{cases}$ **fails IVP** at $x = 1$ (break in graph).



Discontinuous at $x = 1$ (No IVP)

Key Implications

Proposition: If f has IVP on I , then $f(I)$ is an interval.

Monotonicity + IVP $\implies$ Continuity: If f is monotonic on I and has IVP, then f is continuous on I .

Applications:

- Bolzano's Theorem (existence of roots).
- Boundedness of continuous functions on closed intervals.